

Three publicly available prostate tumour whole-exome sequencing (WES) BAMs
SRR10353673 • SRR10353687 • SRR10353710

Source-data verification
ENA MD5 checksum • samtools quickcheck
BAM index verification

Baseline concordance experiment

Stale-index positive-control experiment

Nine deterministic nested downsampling conditions per source BAM
10% • 20% • ... • 90%

27 paired baseline observations
3 source BAMs × 9 nested depth conditions

Mapped-read extraction
samtools flagstat → F
samtools idxstats → I

Baseline discrepancy assessment
Discrepancy measures: $\Delta = F - I$, $|\Delta|$, and $\% \Delta$
Exact-concordance count

Representative validated baseline BAM
SRR10353673 at 10% depth

Prespecified stale-index perturbations
Remove $k = 1,000$ or $10,000$
mapped alignment records
Retain the original BAM index

Mapped-read extraction
samtools flagstat → F
samtools idxstats → I

Positive-control recovery assessment
Expected Δ compared with observed Δ
Residual $R = \Delta_{\text{observed}} - \Delta_{\text{expected}}$